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DEVELOPMENT OF A FIRST PRINCIPLES EQUIVALENT CIRCUIT MODEL FOR A LITHIUM ION BATTERY

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ABSTRACT

This paper examines the development of a linear single particle model that can be used for model based power train simulation, design, estimation and control in hybrid and electric vehicles. The model assumes that the cell consists of spherical particles in each electrode, neglects the electrolyte dynamics, and uses Padé approximations of the particle transcendental transfer functions. The Padé approximated single particle model matches well with the transcendental model, indicating the accuracy of 3^{rd} order Padé approximations for the particle surface concentrations. This model also accurately reproduces the frequency response of a more complex model from the literature, showing that electrolyte diffusion has limited impact on cell impedance. The model also predicts the experimentally measured EIS and moderate($\leq$ 5C) current pulse charge/discharge of a 3.1 Ah Li-ion cell. The explicit form of the impedance allows the development of an equivalent circuit with resistances and capacitances related to the cell parameters.

1 INTRODUCTION

Li-ion batteries are attracting significant and growing interest due to their many applications in hybrid and electric vehicles. Their high energy and high power density render them an excellent option for energy storage. Vehicle applications also require advanced battery management systems(BMS) that ensure safe, longlife, and efficient utilization. Low order models that can accurately capture the battery dynamics enable Computer Aided Engineering of BMS control algorithms. Most BMS are based on equivalent circuit models [1–8] that are easily integrated with the BMS electronics but may not retain the links with the underlying physiochemical processes in the cells. Often, these models are emperical and cannot be used for integrated design of the battery pack and BMS. First principles models capture the essential battery dynamics and have a much better prediction capability compared to the empirical / equivalent circuit models [9–12]. The complexity of these models, however can be a significant barrier to their use in BMS. First principle electrochemical models using porous electrode and concentrated solution theories were developed to study the internal dynamics of a battery [9, 10]. The governing partial differential equations were numerically solved in a computational fluid dynamics framework, making this approach computationally expensive and too slow for real time applications. Ramadass *et al* [13] modified the equations in [9, 10] to incorporate capacity fade. An extensive review of the existing mathematical models for both Li-ion and Nickel battery systems was provided by Gomadam *et al* [14]. In a typical HEV or Plug in HEV, batteries are usually pulse charged and discharged within a relatively narrow state of charge range from 30% to 70%. A reduced order model that has been linearized at 50% SoC range, for example can be sufficiently accurate and low order for model-based BMS. Smith *et al* [11] developed a control oriented model from first principles by analytically and numerically solving the governing partial differential equations to obtain the final impedance transcendental transfer function. Although this model is useful for BMS control and estimation, it cannot be used for parameter identification because the parameters are not explicitly related to the final, polynomial transfer function coefficients.

The single particle (SP) model is a simplified, physics-based model in which each electrode is represented by a single spherical active particle and electrolyte diffusion is neglected. Haran *et al* [15] presented an SP model of a metal hydride system that was adapted to the Li-ion cells [16]. Santhanagopalan *et al* [17]

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compared the accuracy of the single particle model with a more complex pseudo two dimensional model and a simplified porous electrode model using simulated cycling performance and calculation time as metrics. Chaturvedi *et al* [18] also developed an SP model and compared its performance with a full order model for various current inputs. Despite the substantial loss of information due to the SP model simplifications, it can be extremely valuable for control and estimation purposes.

In this paper we combine the impedance based modeling approach developed by Smith *et al* [11] with the SP ideas of [16–18] to develop a linear, control oriented, single particle model of Li-ion cells. The Padé approximation [19, 20] efficiently discretizes the particle diffusion. Frequency and time domain experimental data is used to validate the model. An equivalent circuit realization of the SP impedance transfer function is presented.

2 IMPEDANCE MODEL FORMULATION

The Li-ion cell consists of three domains: the negative composite electrode, separator, and positive composite electrode as shown in Fig.1. Lithium metal oxide ($LiMO_2$) and lithiated carbon (Li_xC) are the active materials in the positive and negative electrodes, respectively. The metal in the positive electrode is a transition metal, typically Co. The active materials are bonded to metal foil current collectors at both ends of the cell and electrically isolated by a microporous polymer separator film or gel-polymer. Liquid or gel-polymer electrolytes enable lithium ions (Li^+) to diffuse between the positive and negative electrodes. The lithium ions insert into or deinsert from the active materials via an intercalation process. The porous separator serves as an electronic insulator, forcing electrons to follow an opposite path through an external circuit or load.

In the positive electrode during charge, the active material is oxidized and lithium ions are de-intercalated as follows

$$\mathrm{Li_{1-x}CoO_2 + xLi^+ + xe^-} \underset{\text{charge}}{\overset{\text{discharge}}{\rightleftarrows}} \mathrm{LiCoO_2}. \tag{1}$$

In the negative electrode during charge, the active material is reduced and lithium ions that migrate from the positive electrode travel through the electrolyte via diffusion and ionic conduction are intercalated in the reaction

$$\mathrm{Li_xC} \underset{\text{charge}}{\overset{\text{discharge}}{\rightleftarrows}} \mathrm{C + xLi^+ + xe^-}. \tag{2}$$

The reactions (1) and (2) are reversible so the battery can be charged and discharged thousands of times.

In the single particle model, each electrode is represented by a single spherical particle of active material and the model is developed under the assumption of negligible electrolyte diffusion. Charge and Li^+ conservation equations in the electrolyte phase [9–11] are not considered here.

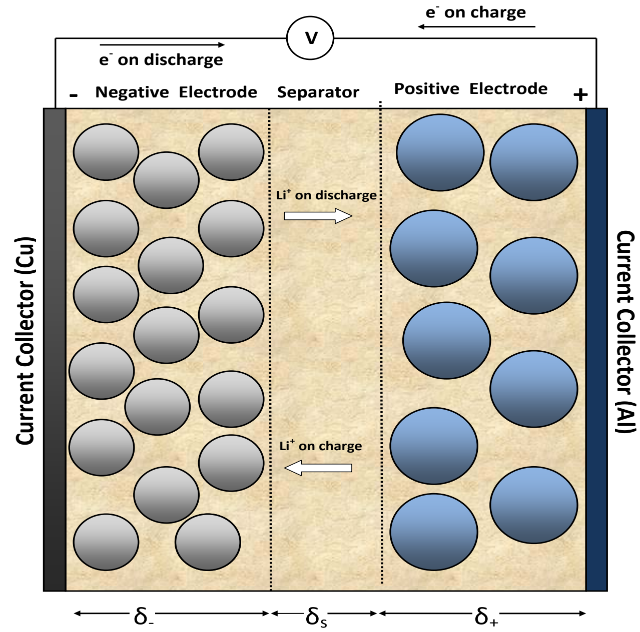

Figure 1. 1D SCHEMATIC OF THE LI-ION BATTERY MODEL.

Conservation of Li in a single spherical active material particle is described by Ficks law of diffusion,

$$\frac{\partial c_s}{\partial t} = \frac{D_s}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial c_s}{\partial r}\right) \quad \text{for } r \in (0, R_s), \tag{3}$$

where $r \in (0, R_s)$ is the radial coordinate, R_s is the particle radius, $c_s(r,t)$ is the concentration of Li^+ ions in the particle as a function of radial position r and time t, and D_s is the solid phase diffusion coefficient. We use the subscripts s, and s,e to indicate solid phase, and solid/electrolyte interface, respectively. The boundary conditions are

$$\left.\frac{\partial c_s}{\partial r}\right|_{r=0} = 0, \tag{4}$$

$$D_s \left.\frac{\partial c_s}{\partial r}\right|_{r=R_s} = -\frac{j}{a_s F}, \tag{5}$$

where $j(x,t)$ is the rate of electrochemical reaction at the particle surface (with $j > 0$ indicating ion discharge), F is Faraday's constant (96487 C/mol) and a_s is the specific interfacial surface area.

For the spherical active material particles occupying electrode volume fraction ε_s, the interfacial surface area is $a_s = 3\varepsilon_s/R_s$.

Taking the Laplace transform of Eq.(3) and solving it with respect to boundary conditions (4) and (5), Jacobsen and West [21] give the solid state diffusion impedance of a spherical active material particle as

$$\frac{C_{s,e}(s)}{J(s)} = \frac{1}{a_s F}\left(\frac{R_s}{D_s}\right)\left[\frac{\tanh(\beta)}{\tanh(\beta)-\beta}\right], \tag{6}$$

where $\beta = R_s\sqrt{s/D_s}$

The Butler-Volmer electrochemical kinetics are

$$j = a_s i_o\left\{\exp\left[\frac{\alpha_a F}{RT}\eta\right] - \exp\left[-\frac{\alpha_c F}{RT}\eta\right]\right\}. \tag{7}$$

where i_0 is the exchange current density, R is the universal gas constant, T is the temperature, α_a and α_c are the anodic and cathodic transfer coefficients, respectively.

The overpotential, η,

$$\eta = \phi_s - U. \tag{8}$$

drives the current flow across the solid/electrolyte interface,where ϕ_s is the solid phase potential. The equilibrium potential,$U(c_{s,e})$, depends on the solid phase concentration at the particle surface.

Linearization of the Butler-Volmer Eq.(7) yields

$$\eta = \frac{R_{ct}}{a_s} j \tag{9}$$

where the charge transfer resistance, $R_{ct} = \frac{RT}{i_0 F(\alpha_a+\alpha_c)}$. Equations (6)-(9) apply to both electrodes but with different parameters.

Voltage across the cell terminals has contributions from both electrodes,

$$V(t) = \phi_s(L,t) - \phi_s(0,t) - \frac{R_f}{A} I(t) \tag{10}$$

where R_f is the contact resistance associated with physical connections to the cell and A is the cell surface area. Substitution of the solutions for ϕ_s in the two electrodes into Eq.(10) results in the impedance transfer function for the single particle model,

$$\frac{V(s)}{I(s)} = \frac{\eta_+(s)}{I(s)} + \frac{\partial U}{\partial c_{s,e+}}\frac{C_{s,e+}(s)}{I(s)} - \frac{\eta_-(s)}{I(s)} - \frac{\partial U}{\partial c_{s,e-}}\frac{C_{s,e-}(s)}{I(s)} - \frac{R_f}{A} \tag{11}$$

where the electrode potentials U are linearized with the assumed constant slope $\frac{\partial U}{\partial c_{s,e}}$.

Using the linearized Butler Volmer equation (9) we have,

$$\frac{\eta(s)}{I(s)} = \frac{R_{ct}}{a_s}\frac{J(s)}{I(s)} = \frac{R_{ct}}{a_s A\delta} \tag{12}$$

where a uniform current density has been assumed across the thickness δ of the electrode.

Similarly, the transfer function $\frac{C_{s,e}(s)}{I(s)}$ can be re-written as,

$$\frac{C_{s,e}(s)}{I(s)} = \frac{C_{s,e}(s)}{J(s)}\frac{J(s)}{I(s)} = \frac{1}{a_s F A\delta}\left(\frac{R_s}{D_s}\right)\left[\frac{\tanh(\beta)}{\tanh(\beta)-\beta}\right], \tag{13}$$

where $\frac{C_{s,e}(s)}{J(s)}$ is the solid state diffusion impedance transfer function defined in Eq.(6).

Substituting Eqs.(12) and (13) to (11) we obtain the overall cell impedance,

$$\begin{aligned}\frac{V(s)}{I(s)} &= \frac{R_{ct+}}{a_{s+}}\frac{1}{A\delta_+} - \frac{R_{ct-}}{a_{s-}}\frac{1}{A\delta_-}\\ &+ \frac{\partial U}{\partial c_{s+}}\frac{1}{A\delta_+}\frac{R_s}{a_s F D_{s+}}\left[\frac{\tanh(\beta_+)}{\tanh(\beta_+)-\beta_+}\right]\\ &- \frac{\partial U}{\partial c_{s-}}\frac{1}{A\delta_-}\frac{R_s}{a_s F D_{s-}}\left[\frac{\tanh(\beta_-)}{\tanh(\beta_-)-\beta_-}\right] - \frac{R_f}{A}\end{aligned} \tag{14}$$

3 DISCRETIZATION

The infinite dimensional single particle model in Eq.(14) includes transcendental transfer functions that are discretized by a Padé Approximation [19, 20]. The Padé Approximation works well in this case because the transcendental transfer functions are infinitely differentiable and can be expanded in a power series at the origin. The N^{th} order Padé approximation of a transfer function *G(s)* is a ratio of two polynomials in s where the denominator is of order N. For a proper transfer function, the numerator is of order N or less. The Padé Approximation Method can produce transfer functions with numerators of order 1 to N but the numerator order should be chosen to best match the high frequency phase asymptote. The high frequency phase asymptotes for the transcendental transfer functions (6) are best matched by a numerator of order N.

We assume that the transfer function can be expanded in a power series at the origin as follows

$$G(s) = \sum_{k=0}^{2(N+1)} c_k\, s^k, \tag{15}$$

where the coefficients c_k are calculated by repeated differentiation and evaluation of $G(s)$ at $s = 0$,

$$c_k = \left.\frac{d^k\, G(s)}{ds^k}\right|_{s=0}. \tag{16}$$

If $G(s)$ has a pole at the origin then we apply the power series expansion to $G^*(s) = s\,G(s)$.

The N^{th} order Padé approximation transfer function

$$P(s) = \frac{\sum_{m=0}^{N} b_m\, s^m}{1+\sum_{n=1}^{N} a_n\, s^n} = \frac{num(s)}{den(s)}, \tag{17}$$

where we assume that the denominator and numerator both have order N. To determine $P(s)$ we must calculate the $N+1$ b_m and N a_m coefficients. The zeroth order term in the denominator is assumed to have a unity coefficient to normalize the solutions. The $2N+1$ linear equations that can be solved for the coefficients are determined from the polynomial equation

$$den(s)\sum_{k=0}^{2(N+1)} c_k s^k - num(s) = 0, \tag{18}$$

where the coefficients c_k are known from the power series expansion. Eq.(18) produces a polynomial of order $2N(N+1)$ in s. The right hand side equals zero for all s so the coefficients must be zero. The first $N+1$ coefficients of s depend on both the unknown a_n and b_n coefficients. The remaining coefficients depend only on a_n. Thus, we set the coefficients of s^{N+2} to s^{2N+1} equal to zero to solve for $a_1,\ldots,a_N$. Then we substitute these solutions $a_1,\ldots,a_N$ into the coefficients of s^0 to s^N and set them equal to zero to solve for $b_0,\ldots,b_N$. A third order Padé approximation is generated for each particle transfer function resulting in the impedance transfer function

$$Z(s) = K + \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s} + \frac{d_2 s^2 + d_1 s + d_0}{s^3 + c_2 s^2 + c_1 s} \tag{19}$$

where the numerator and denominator coefficients are shown in Tab.(1) with

$$C^+ = 21\frac{\partial U^+}{\partial c^+_{s,e}},\; C^- = 21\frac{\partial U^-}{\partial c^-_{s,e}} \tag{20}$$

and K is the total resistance

$$K = \frac{R_{ct_+}}{a_{s_+}}\frac{1}{A\delta_+} + \frac{R_{ct_-}}{a_{s_-}}\frac{1}{A\delta_-} - \frac{R_f}{A} \tag{21}$$

The transfer function (19) depends only on five independent parameters K,

$$\alpha_1 = \frac{C^+}{AFa_s^+ L^+ R_s^+},\; \alpha_2 = \frac{D_s^+}{\left[R_s^+\right]^2},\beta_1 = \frac{C^-}{AFa_s^- L^- R_s^-} \text{ and } \beta_2 = \frac{D_s^-}{\left[R_s^-\right]^2}. \tag{22}$$

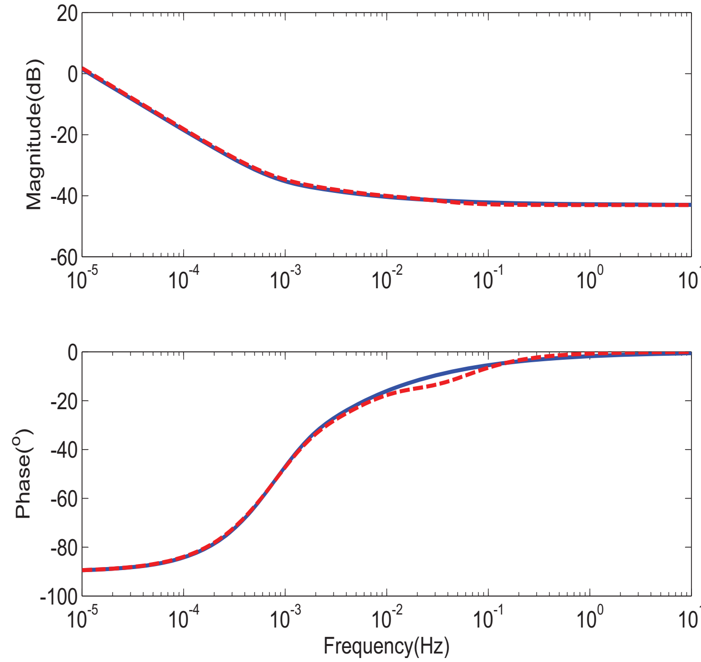

Figure 2. TRANSCENDENTAL (BLUE-SOLID) AND PADÉ APPROXIMATED (RED-DASHED) SP MODEL FREQUENCY RESPONSES.

Table 1. COEFFICIENT VALUES

Numerator	Value	Denominator	Value
b_0	$\frac{495C^+\left[D_s^+\right]^2}{AFa_s^+\delta_+\left[R_s^+\right]^5}$	a_0	0
b_1	$\frac{60C^+D_s^+}{AFa_s^+\delta_+\left[R_s^+\right]^3}$	a_1	$\frac{3465\left[D_s^+\right]^2}{\left[R_s^+\right]^4}$
b_2	$\frac{C^+}{AFa_s^+\delta_+R_s^+}$	a_2	$\frac{189D_s^+}{\left[R_s^+\right]^2}$
d_0	$\frac{495C^-\left[D_s^-\right]^2}{AFa_s^-\delta_-\left[R_s^-\right]^5}$	c_0	0
d_1	$\frac{60C^-D_s^-}{AFa_s^-\delta_-\left[R_s^+\right]^3}$	c_1	$\frac{3465\left[D_s^-\right]^2}{\left[R_s^-\right]^4}$
d_2	$\frac{C^-}{AFa_s^-\delta_-R_s^-}$	c_2	$\frac{189D_s^-}{\left[R_s^-\right]^2}$

because Eq.(19) can be written as

$$Z(s) = K + \frac{\alpha_1 s^2 + 60\alpha_1\alpha_2 s + 495\alpha_1\alpha_2^2}{s^3 + 189\alpha_2 s^2 + 3465\alpha_2^2 s} + \frac{\beta_1 s^2 + 60\beta_1\beta_2 s + 495\beta_1\beta_2^2}{s^3 + 189\beta_2 s^2 + 3465\beta_2^2 s}. \tag{23}$$

Finally the minimal realization of (23) is only fifth order because the two integrators can be combined into one state.

4 RESULTS AND DISCUSSION

Fig.(2) compares the transcendental and Padé approximated frequency responses of the SP model impedance. The Padé approximated model matched well for the desired 10Hz bandwidth. To establish the importance of electrolyte diffusion, the SP model is compared to the linear, control oriented model developed by Smith *et al*. This pseudo-two dimensional model has two independent spatial variables, x (through the thickness) and r (radial through the particles) and is governed by all four conservation equations of Li^+ species and charge in the solid and electrolyte phases. Fig.(3) shows the excellent agreement between the Padé approximated SP model and this model. The explicit form and simplicity of the SP model is advantageous if only impedance is needed. Predicting the distribution of any variable within the electrodes is not possible with the SP model, however. The Padé

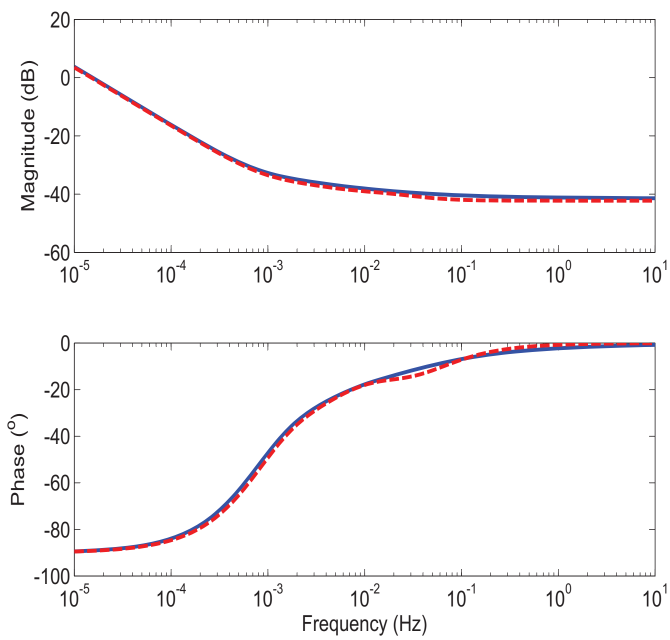

Figure 3. FREQUENCY RESPONSES OF THE MODEL FROM SMITH *ET AL* [11] (BLUE-SOLID) AND THE PADÉ APPROXIMATED SINGLE PARTICLE MODEL(RED-DASHED).

aproximated SP model is also experimentally validated in both the frequency and time domains. Experimental electrochemical impedance spectroscopy(EIS) and pulse train time response data are collected from a commercial battery of capacity 3.1Ah. The SP model parameters are taken from independent tests and manually tuned to match the experimental EIS data. Fig.(4) shows EIS measured at 60% SoC for fresh cells after 1 hour rest following discharge from fully charged state at 1C-rate to 60% SoC. The impedance spectra are obtained with an AC amplitude of 5mV over a frequency range of 0.005 Hz to 50,000 Hz on a Solartron SI 1287 electrochemical interface coupled with Solartron SI 1255B frequency response analyzer. The SP model frequency response extends to lower frequencies not measured experimentally due to equipment and testing time constraints. The experimental data, on the other hand includes frequencies higher than the 10Hz bandwidth of the SP model. Over the frequency range where both results should be accurate, the agreement is quite good.

Pulse discharge and charge at 60% SoC and different C-rates(2C, 5C and 10C) are tested experimentally on an Arbin BT-2000 battery cycler for different pulse durations (2 seconds, 10 seconds and 30 seconds). Fig.(5) shows that the tuned model was validated by comparing the voltage response of the model with the experimental voltage response to an input pulse current. Pulse discharge and charge at 60% SoC and different C-rates (2C, 5C and 10C) were performed for different pulse durations. Fig.(5) shows that the model matches very well with the experiment for the low currents of 2C and 5C but has significant error at the higher current of 10C due to the linearization of the Butler Volmer equation.

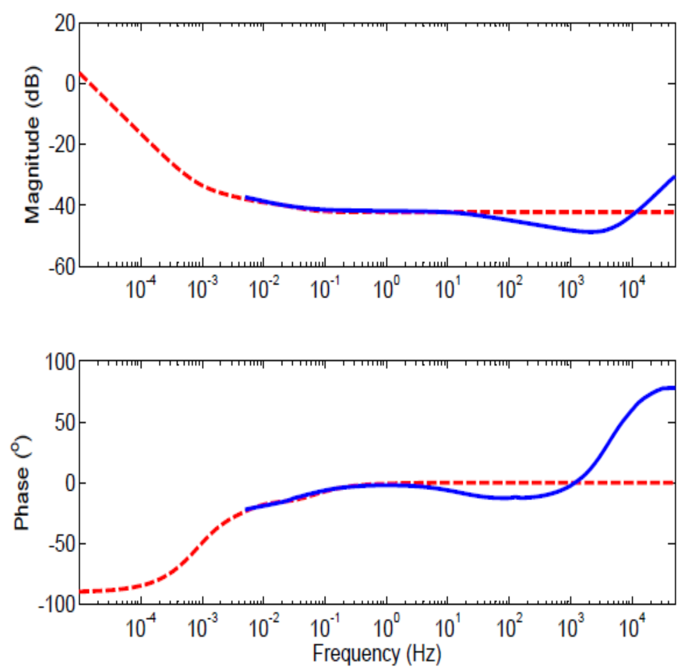

Figure 4. EXPERIMENTAL EIS (BLUE-SOLID) AND PADÉ APPROXIMATED SINGLE PARTICLE MODEL (RED-DASHED) FREQUENCY RESPONSES.

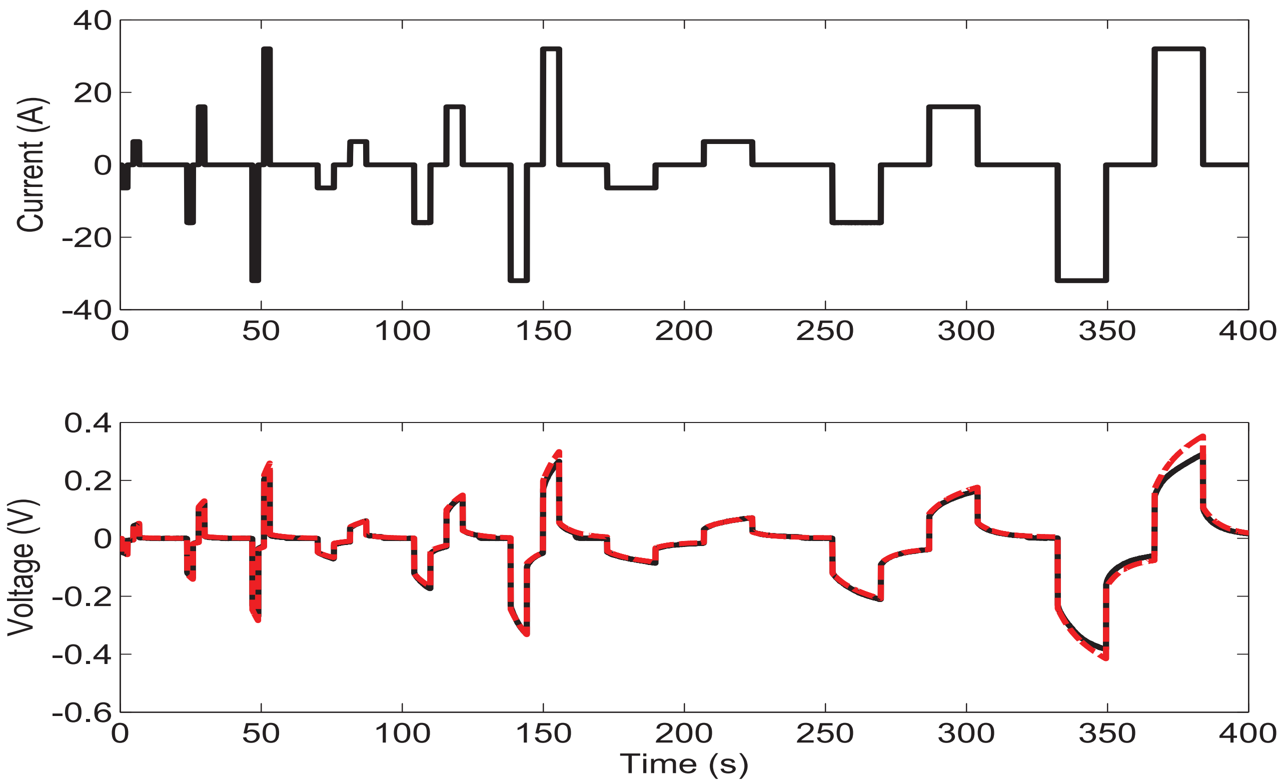

Figure 5. EXPERIMENTAL (BLACK-SOLID) AND SINGLE PARTICLE MODEL (RED-DASHED) PULSE CHARGE/DISCHARGE TIME RESPONSE.

5 FIRST PRINCIPLES EQUIVALENT CIRCUIT MODEL

An advantage of the Padé approximated SP model is that the coefficients of the impedance transfer function are explicit functions of the physical parameters. This low order transfer function can be converted into an equivalent circuit with passive circuit elements like resistors and capacitors with resistances and capacitances that can be physically related to the model parameters. Fig.(6) shows an example equivalent circuit(circuit realization is not unique) that combines four parallel RC circuits with a capacitor and resistor.

The impedance of the equivalent circuit in Fig.(6) is

$$Z(s) = R_1 + \frac{1}{C_1 s} + \frac{\frac{1}{C_2}}{s+\frac{1}{R_2C_2}} + \frac{\frac{1}{C_3}}{s+\frac{1}{R_3C_3}} + \frac{\frac{1}{C_4}}{s+\frac{1}{R_4C_4}} + \frac{\frac{1}{C_5}}{s+\frac{1}{R_5C_5}}, \tag{24}$$

In pole/residue form equating Eq.(24) with the pole residue expansion of the padé approximated SP model transfer function Eq.(23) yields the explicit relationships between resistances and capacitances and physical parameters in Tab.(2)

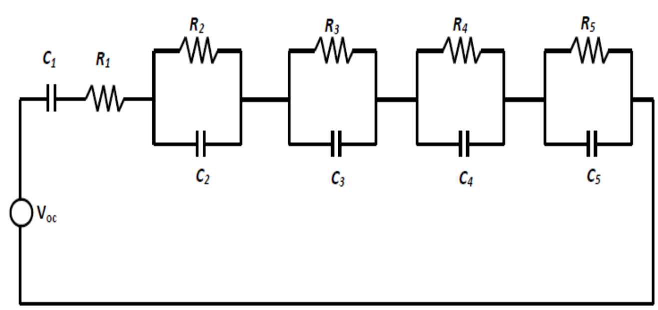

Figure 6. EQUIVALENT CIRCUIT MODEL OF PADÉ APPROXIMATED SINGLE PARTICLE MODEL.

6 CONCLUSIONS

The Padé approximated SP model matches well with the transcendental model, indicating the accuracy of 3^{rd} order Padé approximations for the particle surface concentrations. This model also accurately reproduces the frequency response of a more complex model from the literature, showing that electrolyte

Table 2. CIRCUIT PARAMETERS IN TERMS OF CELL PARAMETERS

Capacitor	Capacitance	Resistor	Resistance
C_1	$\frac{7}{\alpha_1+\beta_1}$	R_1	K
C_2	$\frac{9.6246}{\alpha_1}$	R_2	$\frac{0.0051\alpha_1}{\alpha_2}$
C_3	$\frac{1.3277}{\alpha_1}$	R_3	$\frac{0.0045\alpha_1}{\alpha_2}$
C_4	$\frac{9.6246}{\beta_1}$	R_4	$\frac{0.0051\beta_1}{\beta_2}$
C_5	$\frac{1.3277}{\beta_1}$	R_5	$\frac{0.0045\beta_1}{\beta_2}$

diffusion has limited impact on cell impedance. The model also predicts the experimentally measured EIS and moderate($\leq 5C$) current pulse charge/discharge of a 3.1 Ah Li-ion cell. The explicit form of the impedance allows the development of an equivalent circuit with resistances and capacitances related to the cell parameters.

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